

30.07.20

Coding Challenge

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MLA0407

Multi-layer perceptron:

* A multi-layer perceptron (MLP) is a type of Artificial Neural Network (ANN) used for solving classification and prediction problems.

* It consists of an input layer, one or more hidden layers and an output layer.

* Forward propagation is used to compute the network's predictions.

* Backpropagation updates the weights and biases based on prediction errors.

* The MLP is implemented from scratch using Python and Numpy.

Code:-

```
import numpy as np
```

```
def sigmoid(x):
```

```
    return 1 / (1 + np.exp(-x))
```

```
X = np.array([[25, 30000, 10]])
```

```
X = X / np.max(X)
```

```
np.random.seed(1)
```

$w = \text{np.random.rand}(3,1)$

$b = \text{np.zeros}(1,1)$

$\text{output} = \text{sigmoid}(\text{np.dot}(x, w) + b)$

$\text{print}(\text{"predicted output:"}, \text{output}).$